Marcus Lo

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EDUCATION

University of California, Irvine *Expected Graduation: 06/2027*
Bachelor of Science in Software Engineering
Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, User Interaction Software, Human Computer Interaction, Intro to Python Learning Assistant, Computer Organization, Software Testing & Quality Assurance

EXPERIENCE

Moody's Analytics *06/2026 - 08/2026*
Software Engineering Intern

- Rebuilt a cross-environment validation pipeline for a property-intelligence API, replacing manually maintained API configuration with automatic feature and datapack discovery while validating 38 datapack versions, 200+ features, and approximately 2,800 API requests per sweep using Python, asyncio, aiohttp, and Pydantic
- Surfaced cross-environment API discrepancies that single-environment checks could miss, including requests succeeding in staging but failing in integration, through feature-level response pairing and deep JSON comparison; backed the pipeline with 76 unit tests
- Benchmarked up to 19x higher throughput in testing by designing decoupled per-environment queues with 20 concurrent workers

AIQUANTA *08/2025 - 09/2025*
Software Engineering Intern

- Engineered Python-based autonomous drone control system for DJI Tello, enabling 3 dimensional stabilization and face tracking using PID and OpenCV AI to center and follow user's face automatically with <25ms delay
- Implemented swarm functionality, controlling several drones simultaneously with <50ms delay using a design that supports upward scalability
- Wrote documentation and procedures for connecting, configuring, and programming drones

AIphotonics *08/2023 - 09/2023*
STEM Education Intern

- Led STEM robot kit development, creating content that was used by 150 high school students
- Wrote code and designed lesson plans that generated over $65,000 in revenue with minimal working costs
- Created documentation and instruction manuals, collaborated with coworkers and CEO on various software projects that built upon previous interns' work

PROJECTS

Curtain Call Alarm Clock C++, Swift, Arduino, GitHub

- Engineered IoT alarm system integrating Swift iOS app with Arduino Nano hardware to automate curtain opening
- Developed method in Swift application to send bluetooth signal asynchronously to embedded platform
- Reduced power consumption by 90% using BLE in Arduino C++ integrated with HC-08 sensor

Touch Sensitive Electric Piano C++, Python, Arduino, GitHub

- Built real-time MIDI piano system in C++ and Python with Arduino-based touch sensors, integrating FluidSynth library for audio synthesis and custom serial communication protocol
- Implemented cross-platform MIDI input/output handling by connecting Arduino hardware via USB serial to Python and C++ applications, utilizing pyserial to enable robust sensor interfacing, MIDI event handling, and synthesized audio output

SKILLS

- **Programming Languages:** Python, C++, Java, SQL, HTML, CSS, Swift
- **Tools:** Git, GitHub, Jupyter Notebook, Microsoft Office, Figma, asyncio, Arduino, RESTful API, Jenkins
- **Skills:** Backend, API integration, Embedded Systems, Software Testing & Debugging, Agile Methodology